

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) — Semester II Examination, 2023-24

**Paper No. MTB-202: Statics & Dynamics**

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 074904

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

**Question 1.** (Compulsory — 2 marks each.)

- (a) Obtain a single force or couple equivalent to four forces with components  $(2, 3)$ ,  $(3, -1)$ ,  $(-4, -4)$ ,  $(-1, 2)$  acting at  $(0, 1)$ ,  $(1, -1)$ ,  $(2, 3)$ ,  $(-1, -2)$  respectively.
- (b) Find the work done by the thrust at the ends of a rod when displacement alters its length.
- (c) Prove that a system of coplanar forces is in equilibrium if the algebraic sum of moments about each of three non-collinear points is zero.
- (d) A particle of mass  $m$  moves along a path with linear velocity  $v$  at distance  $r$  from the origin. Obtain the angular velocity  $\omega$ .
- (e) Which of the following are NOT properties of SHM: (i) acceleration always directed to a fixed line; (ii) constant distance from a fixed point; (iii) constant linear velocity; (iv) motion always in a straight line.
- (f) If radial and transverse velocity components are  $\lambda r$  and  $\mu r^2$  ( $\lambda, \mu$  constants), find the polar equation of the path.
- (g) State Kepler's third law in mathematical form and prove it.
- (h) A particle describes a cardioid  $ap^2 = r^3$  under a central force. Find the law of force.
- (i) Is areal velocity of a particle under central force always proportional to angular momentum per unit mass? Give reasons.
- (j) A particle of mass 3 units moves up the inner surface of a smooth vertical circular path of radius 5 with initial velocity 15. Find the reaction at angle  $60^\circ$  ( $g = 9.8$ ).
- (k) A frame has linear motion with acceleration. Is it a Newtonian frame of reference? Give reasons.

**Question 2.**

- (a) Show that a system of coplanar forces can be reduced to a single force through a given point and a couple. [6]

- (b) A system of forces in the plane of triangle  $ABC$  is equivalent to a single force at  $A$  along the internal bisector of  $\angle BAC$  and a couple  $G_1$ . If moments about  $B$  and  $C$  are  $G_2$  and  $G_3$ , show  $(b + c)G_1 = bG_2 + cG_3$ , where  $b = CA$ ,  $c = AB$ . [6]

**Question 3.**

- (a) State and prove the principle of virtual work for a system of coplanar forces. [6]
- (b) A heavy elastic string of natural length  $2\pi a$  is placed round a smooth cone of semi-vertical angle  $\theta$ , axis vertical. Weight  $W$ , modulus  $\lambda$ . Prove it is in equilibrium when it forms a circle of radius  $a\left(1 + \frac{W \tan \theta}{2\pi \lambda}\right)$ . [6]

**Question 4.**

- (a) Derive the tangential and normal components of acceleration of a particle moving in a plane curve. Also write the radial and transverse components. [6]
- (b) A point moves in a curve so that tangential and normal accelerations are equal and the tangent rotates with constant angular velocity. Show the intrinsic equation is  $s = Ae^\theta + B$ . [6]

**Question 5.**

- (a) Show that in SHM the average speed and average acceleration are obtained by multiplying their maximum values by  $\frac{2}{\pi}$ . [6]
- (b) Discuss the motion of a particle in a frame rotating with constant angular velocity about a fixed frame and obtain the fictitious forces. [6]

**Question 6.**

- (a) If the central orbit is  $r^3 = a^3 \cos 3\theta$  under force toward the pole, find the law of force and velocity at any position. [6]
- (b) A particle under central acceleration  $\mu/(\text{distance})^2$  is projected with velocity  $V$  at distance  $R$ . Show its path is a rectangular hyperbola if angle of projection is  $\sin^{-1}\left\{\mu/(VR\sqrt{V^2 - 2\mu/R})\right\}$ . [6]

**Question 7.**

- (a) A particle moves on a smooth vertical circular wire of radius  $a$ , projected from the lowest point with velocity  $\sqrt{4ga}$ . Show the reaction is zero after time  $\sqrt{a/g} \log(\sqrt{5} + \sqrt{6})$ . [6]

- (b) Two particles drop from the cusp of a cycloid at interval  $t$ . Prove they meet after time  $2\pi\sqrt{a/g} + t/2$ . [6]